

TinyLlama LoRA Fine-Tuning using PEFT

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Abstract

This project demonstrates parameter-efficient fine-tuning of the TinyLlama 1.1B autoregressive language model using LoRA adapters and PEFT techniques under constrained GPU environments. The model was trained on multiple lightweight text datasets using Google Colab T4 GPUs and tracked using Weights & Biases (W&B). The project focuses on reproducibility, modularity, and efficient training workflows.

1 Introduction

Large Language Models (LLMs) require significant computational resources for full fine-tuning. Parameter-Efficient Fine-Tuning (PEFT) methods such as LoRA (Low-Rank Adaptation) reduce memory and compute requirements by training only a small subset of additional parameters.

This project explores LoRA fine-tuning on TinyLlama 1.1B using lightweight datasets compatible with limited GPU environments like Google Colab T4 GPUs.

2 Model

The selected model was:

- TinyLlama/TinyLlama-1.1B-Chat-v1.0
- 1.1 Billion Parameters
- Transformer-based autoregressive architecture
- Hugging Face compatible
- PEFT + LoRA compatible

The model was loaded using 4-bit quantization through BitsAndBytes for reduced GPU memory usage.

3 Datasets

The following datasets were used:

- TinyStories
- WikiText-2
- AG News
- DailyDialog
- SQuAD

Datasets were converted into text-only format and tokenized using the TinyLlama tokenizer with maximum sequence length 128.

4 LoRA Configuration

Parameter	Value
Rank (r)	8
Alpha	16
Dropout	0.05
Target Modules	q-proj, v-proj

Table 1: LoRA Hyperparameters

5 Training Setup

Training was performed on Google Colab using NVIDIA Tesla T4 GPUs.

Parameter	Value
Batch Size	1
Gradient Accumulation Steps	4
Learning Rate	2e-4
Epochs	1
Max Sequence Length	128
Optimizer	paged_adamw_8bit
Precision	FP16

Table 2: Training Configuration

6 Experiment Tracking

All experiments were tracked using Weights & Biases (W&B).

Tracked Metrics:

- Training Loss
- Learning Rate
- Gradient Norm
- Runtime Metrics
- Global Step

7 Results

7.1 Training Loss



Figure 1: Training Loss Curve

7.2 Learning Rate Schedule

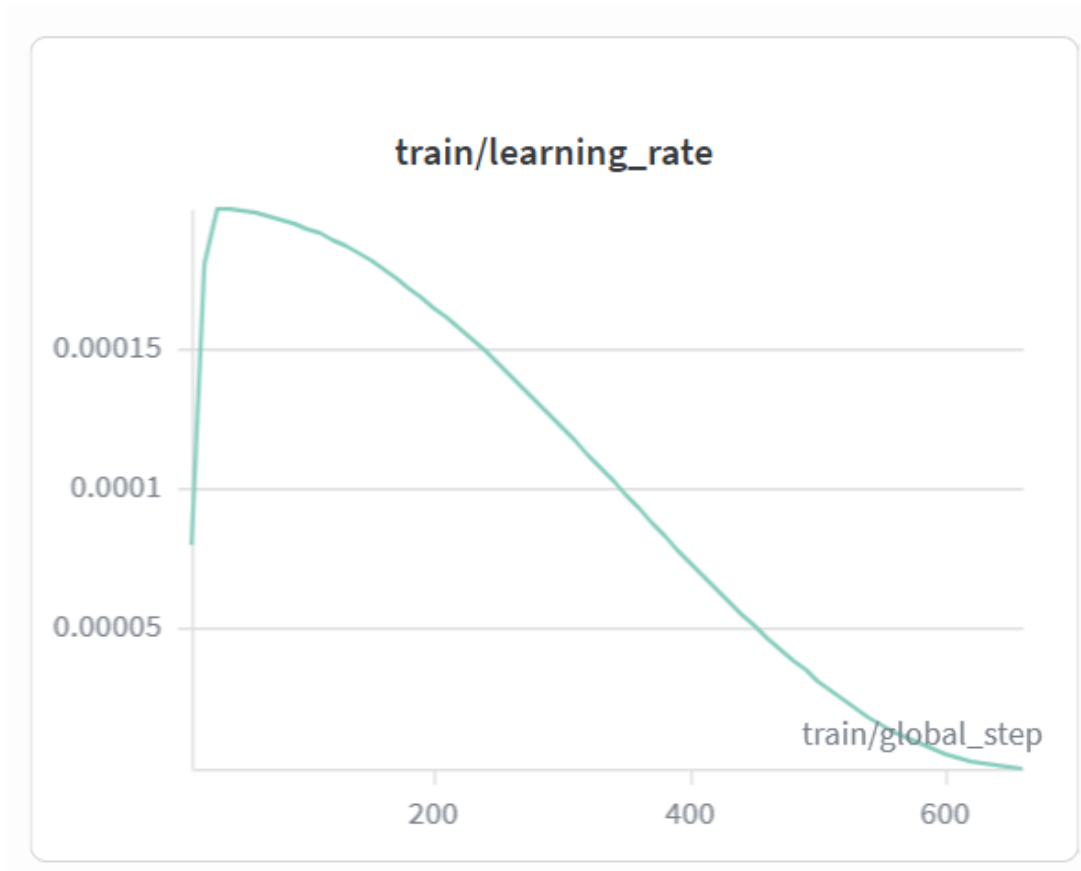


Figure 2: Learning Rate Schedule

7.3 Gradient Norm

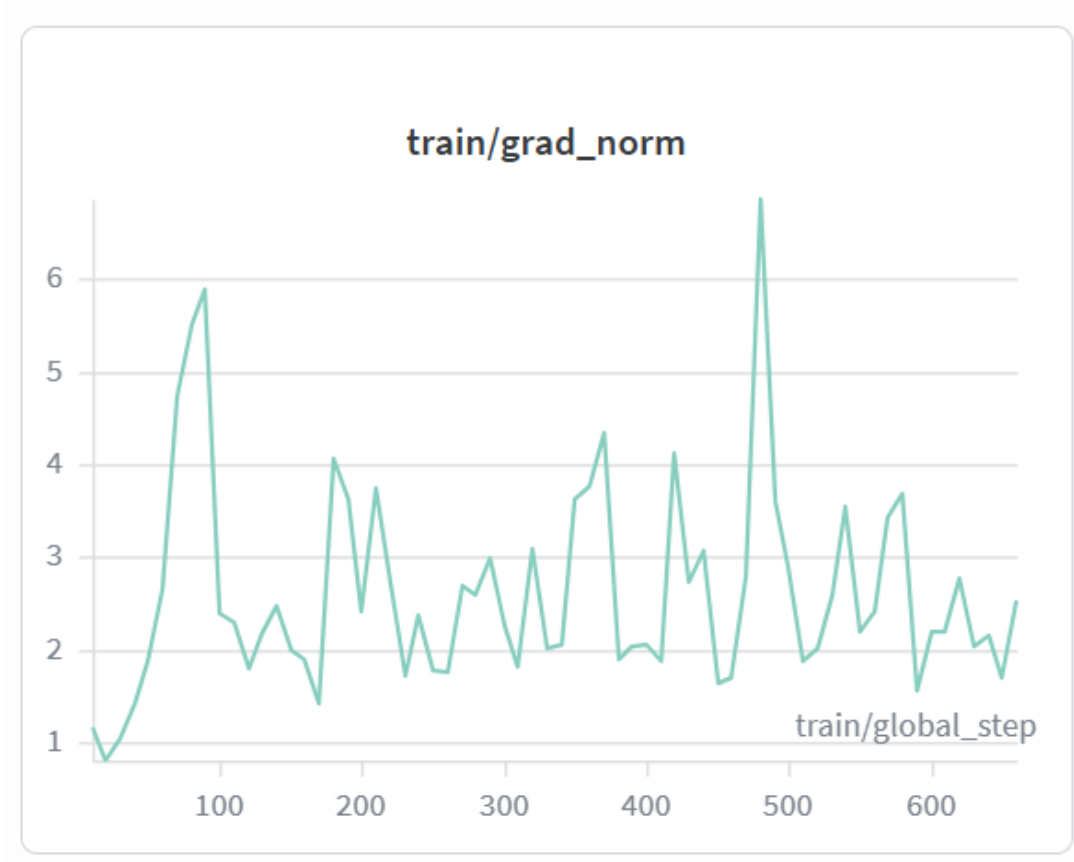


Figure 3: Gradient Norm during Training

8 Inference Performance

Metric	Value
Inference Time	6.09 sec
Tokens Generated	54
Tokens/sec	8.86
Final Training Loss	2.02

Table 3: Inference Metrics

Generated text quality improved slightly after LoRA fine-tuning, though limited training epochs and constrained datasets reduced final coherence.

9 Challenges Faced

- Google Colab GPU usage limits
- Limited VRAM on T4 GPU

- Runtime disconnects
- Dataset compatibility issues
- Managing memory during quantized training

10 Conclusion

This project successfully demonstrated efficient LoRA fine-tuning of TinyLlama 1.1B using PEFT methods under constrained GPU environments.

The workflow achieved:

- Successful LoRA adaptation
- Efficient 4-bit quantized training
- Experiment tracking with W&B
- Modular and reproducible repository setup

Future work may include:

- Larger datasets
- Multi-epoch fine-tuning
- Better evaluation metrics
- Improved inference quality

11 References

- Hugging Face Transformers Documentation
- PEFT Documentation
- TinyLlama Model Card
- Weights & Biases Documentation
- BitsAndBytes Quantization Documentation